

Assignment 3

BMED 8813 BDV

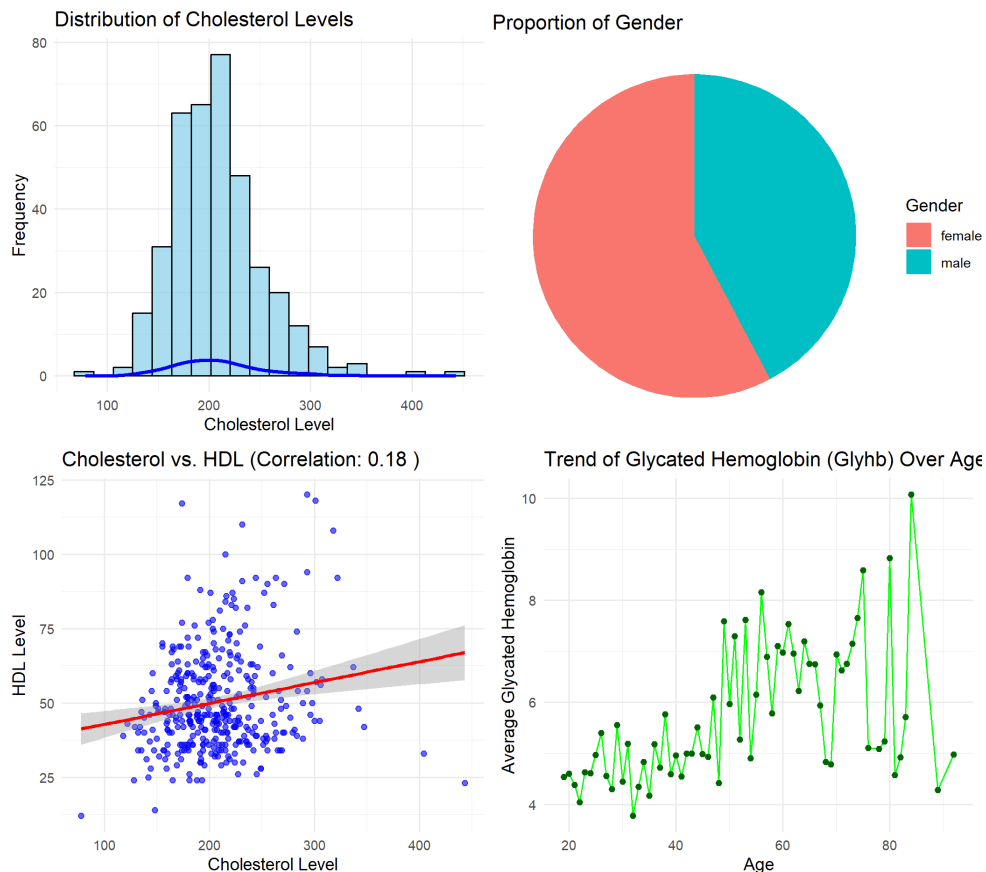
5% of final grade

Due 03/13/2025

1. Figure Design Principles

For this question, please use the [diabetes.csv](#) provided on Canvas. You may use Python, Tableau, or R to create these visualizations. Please base your designs on the information presented in the Week 7 Lecture.

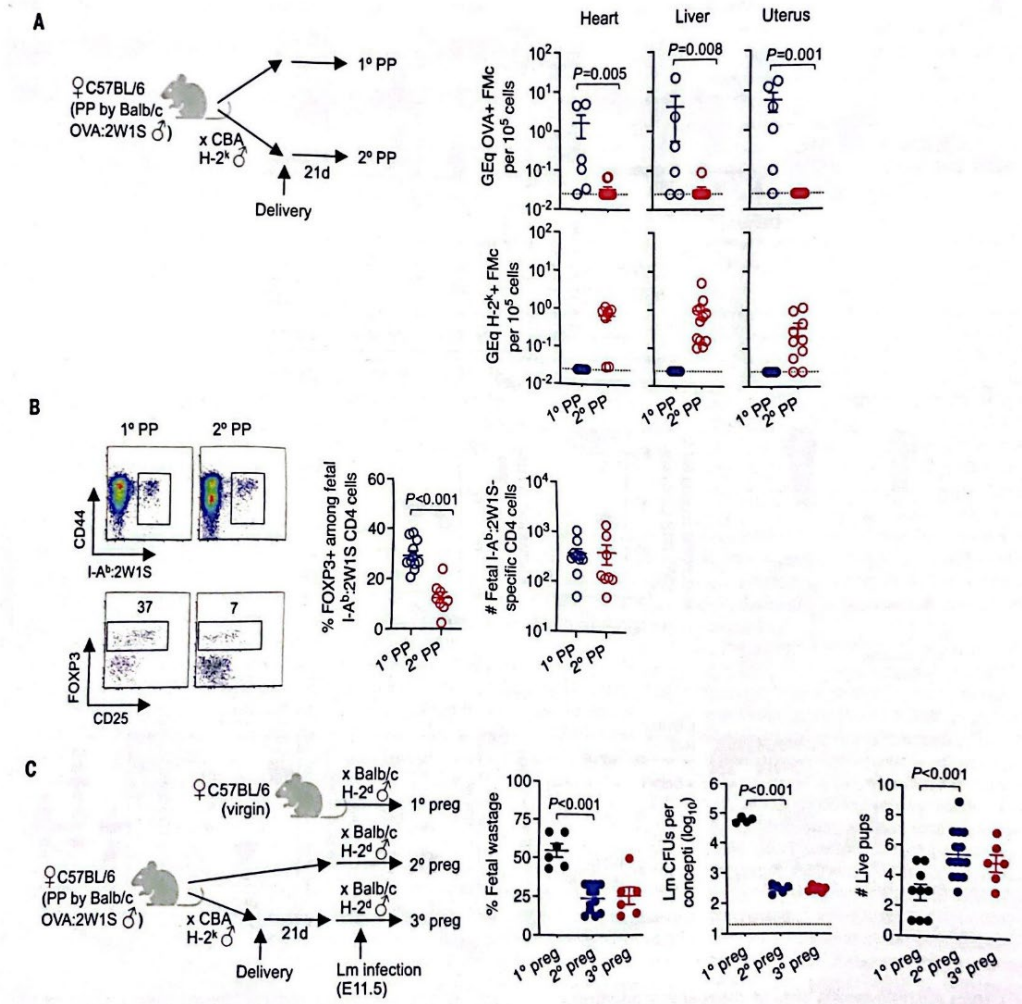
- DISTRIBUTION** - Using that data, please create a histogram, bar chart, or density plot that allows for a clear distribution of one of the variables.
- PROPORTIONS** - Using the data provided, create a proportions-based graph. You may do any other graph, but we would suggest a pie chart.
- ASSOCIATIONS** - Create a scatterplot of two variables. Please include a line of best fit and the coefficient correlation for the line. The line can be non-linear.
- TRENDS** - Please create a graph that allows for the visualization of a variable trend over age.
- HEATMAP** – Please generate a heatmap of chol, stab.glu, hdl, ratio, glyhb (y axis) as an increasing “age” of the patients in the x axis.
- RIDGELINE** – Please create a ridgeline plot using the “location” column for distributions of chol, stab.glu, hdl, ratio, and glyhb using separate graphs for each.



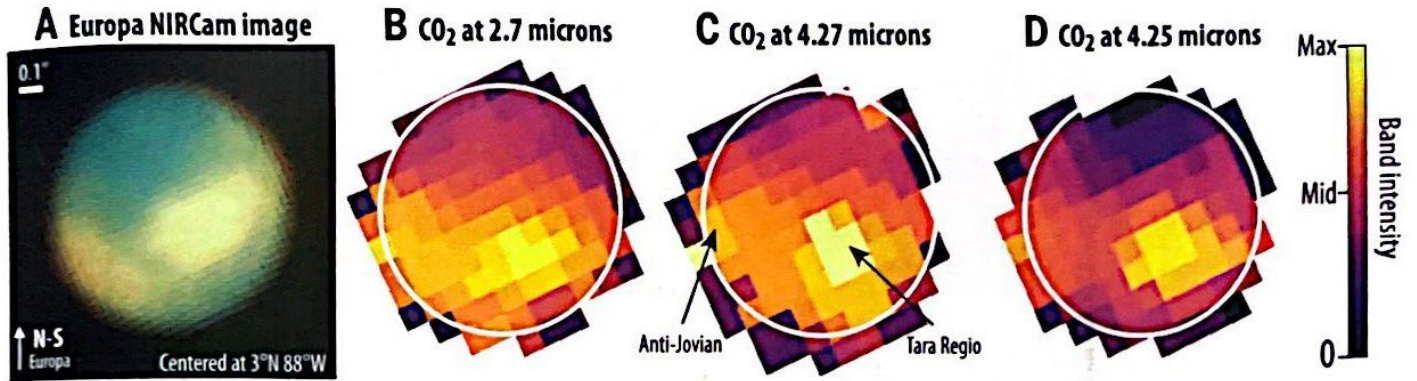
2. Figure Analysis and Data Viz Interpretation

Based on what you have learned in lectures, please critique these **real-world (i.e., scientific publications) figures** and describe how they can be improved. Each section has a given topic of discussion to lead your answer.

a. Please provide critiques on the **usage of space** in this figure.

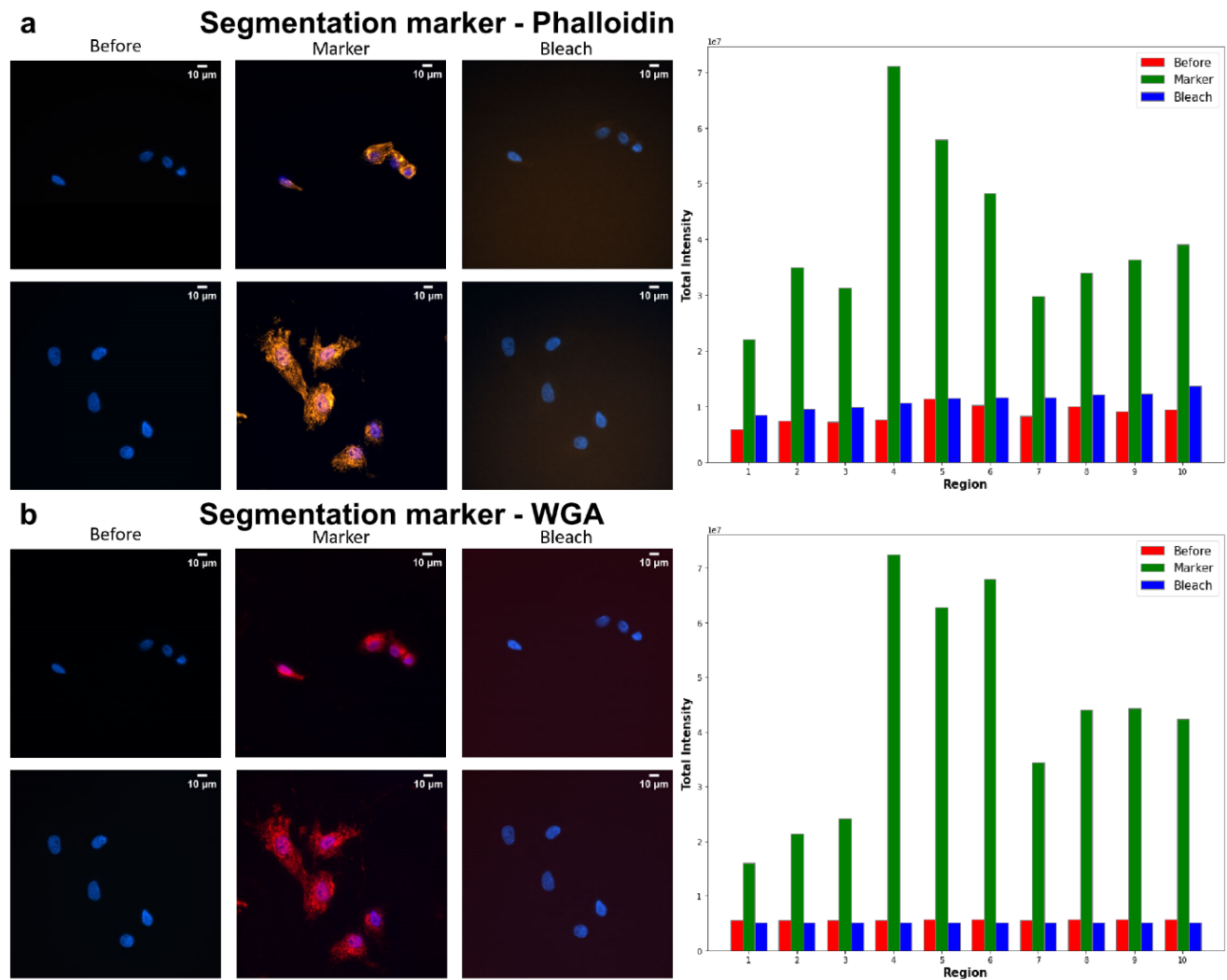


b. Please critique the **color balance and quantification use** of this figure.



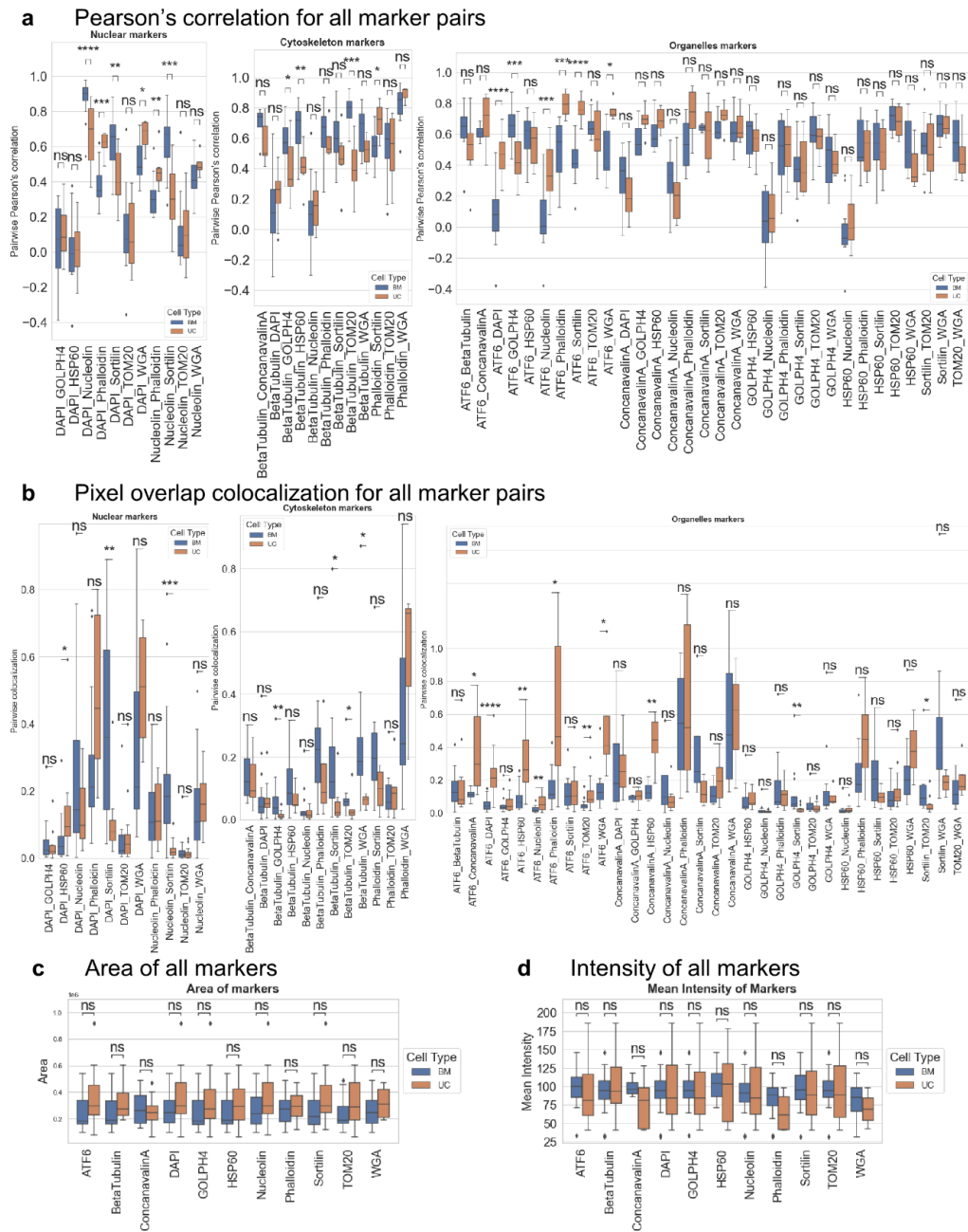
Reference: <https://www.science.org/doi/10.1126/science.adg4270>

c. Please critique the **color accessibility** and the **use of space** in this figure.



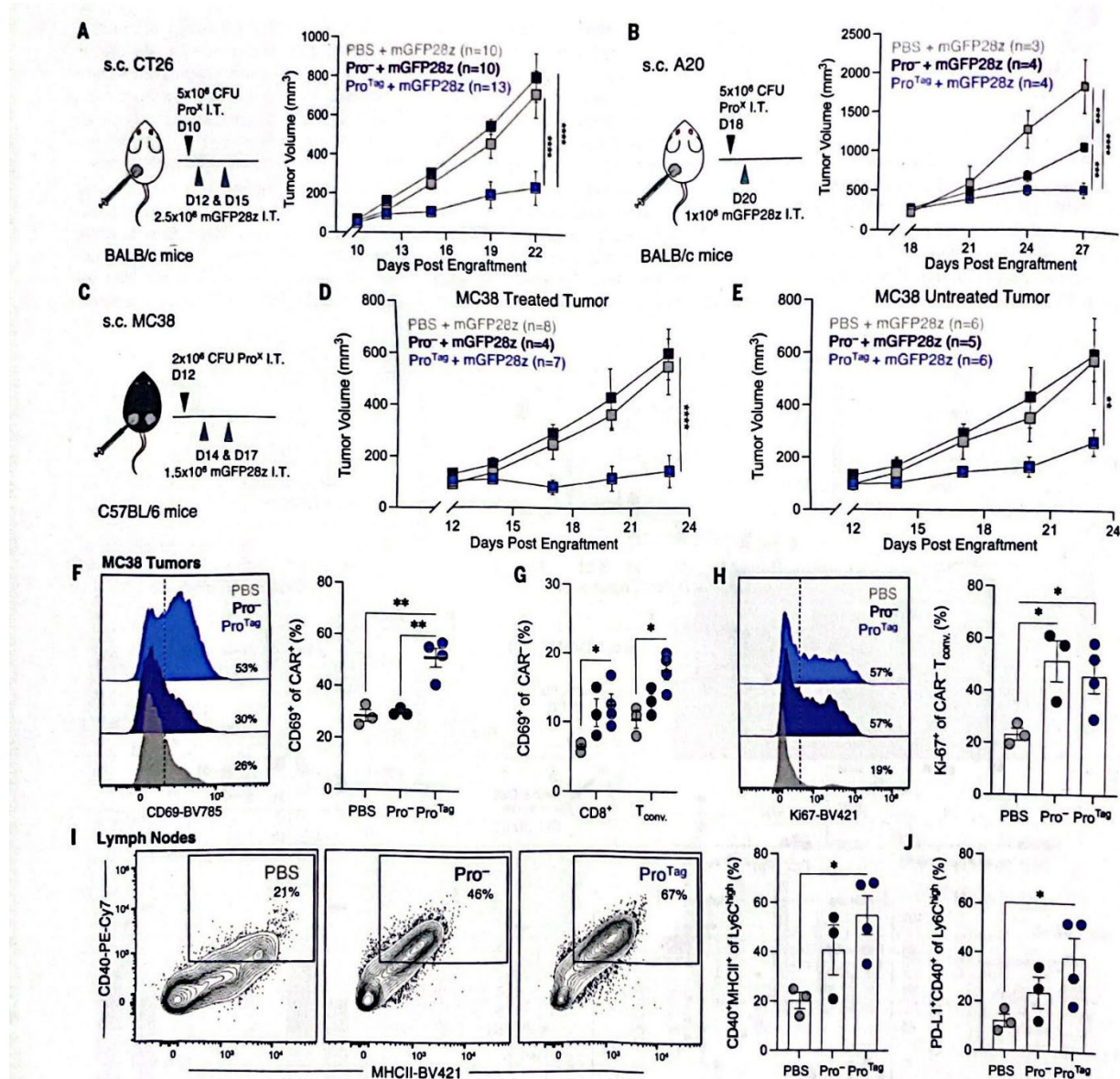
Reference: <https://www.nature.com/articles/s41598-023-32474-y>

d. Please critique the **typography** of this figure.



Reference: <https://www.nature.com/articles/s41598-023-32474-y>

e. Please critique the **color complexity** and **statistics visuals** of this figure.



Reference: <https://www.science.org/doi/10.1126/science.add7034>